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Shape from polarization via ultra-low-sampling single-pixel imaging with decoupled spatial and polarimetric dimensions

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HIGHLIGHTS

- A novel SfP system with decoupled spatial and polarimetric dimensions is proposed, enabling high-precision surface normal imaging at an ultra-low sampling rate of 0.39%.
- An efficient HyCGA-Transformer fusion framework is designed to adaptively focus on key features and enhance feature utilization via cross-scale semantic alignment.
- The proposed method exhibits strong robustness in static scenes, providing a new technological pathway for 3D imaging on resource-limited platforms in static environments.

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ABSTRACT

Shape from Polarization (SfP), a passive three-dimensional (3D) imaging method with high precision, long detection range, simplicity, and low cost, holds significant application value in surface defect inspection, virtual reality, and robotic vision. However, traditional SfP techniques face high storage demands, large bandwidth transmission, and massive data processing burdens, leading to high computational load, energy consumption, and cost, which severely limit their mobile deployment. To address this, we innovatively propose a single-pixel ultra-low sampling SfP system based on decoupled spatial and polarimetric dimensions, combined with a deep fusion network. It acquires raw target data at an ultra-low sampling rate (0.39%) via deep undersampling. Experimental results demonstrate that, at a pixel resolution of 1024×1024, while maintaining reconstruction accuracy comparable to conventional methods, the raw per-frame data volume is reduced from 4.00 MiB of traditional full-sampling with four polarization states to only 0.0156 MiB. Correspondingly, the storage occupation and transmission bandwidth are reduced by 256 times, greatly alleviating the pressure of data storage and transmission at the acquisition terminal. This low-data-volume, low-cost approach effectively addresses the technical bottlenecks of rapid image data transmission and processing in bandwidth-constrained imaging scenarios, thereby offering a new technological pathway for 3D imaging on resource-limited platforms in static environments.

1. Introduction

Shape from polarization (SfP), as a passive optical 3D imaging technique, has become a core technology in fields such as virtual reality, industrial inspection, and robotic perception due to its non-contact nature, high accuracy, and capability for full-surface detail reconstruction [1]. Traditional Stokes vector-based SfP imaging provides a passive solution for 3D reconstruction of complex scenes by extracting surface normals and 3D geometric features of targets from captured information across different polarization states [2]. However, as applications evolve toward dynamic, mobile, and ubiquitous deployment, existing methods face three major technical barriers in data acquisition, transmission, and processing. First, the parallel acquisition of four-polarization-state images demands substantial storage space, leading to high energy consumption and transmission delays on mobile platforms [3]. Second, physics model-based SfP reconstruction methods rely on strict material

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reflectance assumptions, making them susceptible to π-ambiguity and zenith angle errors in complex multi-object scenarios [4,5]. Third, although deep learning-driven SfP approaches improve robustness, their reliance on full-sampling polarization image processing still suffers from high computational complexity and model redundancy, hindering real-time performance [6].

Early research on SfP focused on analytical derivations based on the Fresnel equations and polarized reflection models, establishing surface normal estimation frameworks under idealized assumptions of either pure specular or pure diffuse reflection [7,8]. These methods achieved high accuracy in controlled laboratory environments, such as scenes with single material types and fixed lighting, but faced significant challenges in real industrial applications. In particular, the interplay of specular reflections from metal surfaces and diffuse reflections from plastic materials, together with multi-source illumination interference, often led to considerable errors in zenith-angle estimation [9]. To overcome the limitations of physical models, researchers attempted to incorporate multi-view geometric constraints [10] or multi-sensor fusion (e.g., combining SfP with depth cameras in SfP-D [11]). However, complex imaging conditions and stringent sensor calibration requirements greatly restricted their application in outdoor dynamic scenarios.

The introduction of deep learning has significantly advanced SfP technology [12]. Current methods improve reconstruction accuracy in complex scenes by integrating multi-physical features—such as the interaction between shading cues and specular reflection characteristics, and the fusion of polarization and geometric constraints [13]. Nevertheless, three core challenges remain: At the algorithmic level, multi-scale fusion mechanisms for multi-modal features have yet to be achieved, with insufficient decoupling strategies for physical constraints such as azimuthal π-ambiguity and zenith angle inaccuracies in polarization information [14]. On the data side, reliance on full-sampling polarimetric images leads to high bandwidth demands, limiting real-time processing on mobile devices. In terms of computational efficiency, the extensive parameterization (tens of millions) and high latency associated with complex network architectures hinder their applicability in edge computing. How to construct lightweight, low-data-dependent deep learning frameworks while achieving deep integration of physical priors and data-driven approaches has become a critical direction awaiting a breakthrough in the field.

To alleviate the data burden in SfP tasks, recent studies have explored lightweight imaging strategies based on compressed sensing [15], typically using computational imaging frameworks such as coded apertures [16,17], random masks [18], and single-pixel imaging [19] to reconstruct intensity or polarization images from a limited number of measurements. However, most existing methods focus on compressed reconstruction at the two-dimensional image level [20], without jointly considering compressed sampling, synchronous acquisition of four polarization states, and surface normal recovery for SfP-based 3D reconstruction. In contrast to conventional strategies that compress polarization images as an independent preprocessing step [21], this work embeds the SPI undersampling mechanism into the full SfP reconstruction pipeline, reducing the resource cost of four-polarization-state data acquisition, transmission, and backend processing from the source.

Based on this, this study pioneers the integration of the deep undersampling advantage of single-pixel imaging (SPI) with SfP technology, and proposes an SPI ultra-low-sampling SfP architecture based on decoupled spatial and polarimetric dimensions. Through synergistic innovation in optical encoding and deep learning, three major breakthroughs have been achieved: (1) An SfP system based on SPI ultra-low-sampling with decoupled spatial and polarimetric dimensions is proposed, which maintains high-precision surface normal reconstruction capability even at a sampling rate of 0.39%; (2) A HyCGA-Transformer deep fusion framework is designed, which achieves adaptive focusing on key features by dynamically generating pixel-wise and channel-wise spatial importance maps, and significantly improves feature utilization through a novel cross-scale semantic alignment mechanism; (3) The method demonstrates strong robustness in validation experiments conducted in real static scenes, and this study may offer a new technical pathway for bandwidth-constrained static-scene 3D imaging.

2. Methods

The proposed SPI ultra-low-sampling SfP framework with decoupled spatial and polarimetric dimensions consists of three main components. The first component performs the encoding of spatial and polarimetric information, as well as the acquisition and storage of the encoded measurements using a four-polarization-state quadrant detector. The second component computes polarization prior information from the measured signals. The third component recovers 3D information through a deep learning model based on the derived polarization priors.

2.1. Ultra-low-sampling SPI system based on decoupled spatial dimensions

The designed ultra-low-sampling SPI system based on decoupled spatial dimensions primarily consists of a camera lens, a digital micromirror device (DMD), a four-polarization-state parallel quadrant detector, and a computer. As shown in Fig. 1, the camera lens is responsible for collecting and converging optical signals from the target scene. The DMD performs spatial light modulation on incident light using predefined Hadamard light-field encoding patterns [22]. The four-polarization-state parallel quadrant detector measures the polarized optical signals modulated by the DMD [23,24], while the computer performs signal computation and deep learning-based SfP processing. The specific work-

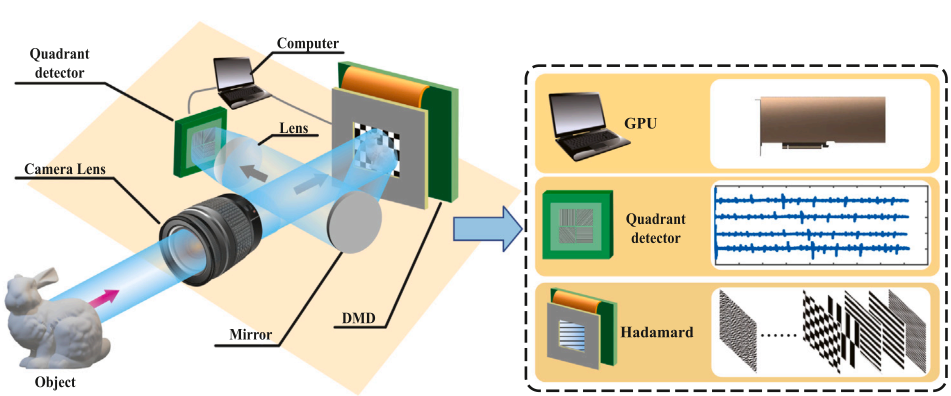


Fig. 1. Schematic diagram of the SPI ultra-low-sampling SfP system based on decoupled spatial and polarimetric dimensions.

flow is as follows: optical signals from the target scene are converged by the camera lens onto the DMD surface [25]. The DMD modulates and encodes the light field via mirror reflection (at a $\pm 12^{\circ}$ output angle), after which the modulated light is focused by a converging lens onto the four-quadrant parallel single-pixel polarization detector. Finally, the four-polarization-state information is obtained in the digital domain through decoupling of the spatial dimensions. The four sensing elements of the four-quadrant parallel single-pixel detector are physically separated. However, each channel records an integrated intensity signal under the same DMD modulation pattern, rather than forming a pixel-resolved image. Therefore, this separation does not introduce pixel-level parallax or spatial misregistration as in conventional array-camera-based SfP systems.

The spatial modulation by the DMD and the integrated measurements from the four-quadrant parallel single-pixel detector together constitute the SPI acquisition scheme used in this work. Compared with SPI schemes based on random masks [18] or Fourier patterns [26], we adopt Hadamard coding with cake-cutting ordering [27] because Hadamard patterns are orthogonal and binary, making them well suited for high-speed DMD projection. The cake-cutting order samples low-frequency structural components first, which helps preserve the object outline and polarimetric structure more effectively than random ordering at low sampling rates. As shown in Fig. 1, the optical information of the target object $O(x, y)$ is converged by an optical lens onto the DMD. The DMD performs spatial light encoding by loading a predefined set of M optical modulation patterns $H^m(x, y)$ (where $m = 1, 2, \ldots, M$) [28]. The encoded information is then acquired in parallel by a four-polarization-state quadrant detector, which captures the four polarization states simultaneously. The measurement signal $B_k^{(m)}$ of the detector for the polarization state k (where $k = 0, \pi/4, \pi/2, 3\pi/4$) is given by Eq. (1):

$$B_k^{(m)} = \iint H^m(x, y) \cdot O(x, y) dx dy. \tag{1}$$

Finally, the polarization image information $G_k(x, y)$ for the $k - th$ polarization state can be obtained by Eq. (2):

$$G_k(X, Y) = < H^{(m)}(x, y) B_k^{(m)} >, \tag{2}$$

where $m = 1, 2, 3, \ldots, M$, represents the number of measurements. $< \cdot > = \frac{1}{M} \sum_{m=1}^{M} (\cdot)$.

In practice, the SPI model is integrated with the training data and serves as a physically interpretable preprocessing stage before the neural network, thereby improving convergence and generalization. As shown in Fig. 2, the acquired measurements are first processed according to the SPI principle to obtain preliminary feature representations. To achieve ultra-low-sampling acquisition, the Hadamard basis is reordered using a cake-cutting sequence [27]. Specifically, for a polarization-state image $O \in \mathbb{R}^{H\times W}$, the image is first reshaped into $O_{2^N} \in \mathbb{R}^{2^N\times 2^N}$ and then vectorized as $O_{2^{2N}} \in \mathbb{R}^{2^{2N}\times 1}$. A cake-cutting-ordered Hadamard light field $H \in \mathbb{R}^{2^{2N}\times 2^{2N}}$ is then selected. Given M measurements, the undersampled light field is denoted by $H_M \in \mathbb{R}^{M\times 2^{2N}}$. After modulation by H_M, the corresponding undersampled polarization measurements are obtained as $B = H_M \times O_{2^{2N}} \in \mathbb{R}^{M\times 1}$. The undersampled polarization information is then reconstructed by second-order correlation as $G = B^T \times H_M \in \mathbb{R}^{1\times 2^{2N}}$. Finally, G is reshaped into $G_{2^N} \in \mathbb{R}^{2^N\times 2^N}$ to produce the undersampled polarization image. The sampling rate is defined as $\mu = \frac{M}{2^N\times 2^N}$. For example, when the image resolution is

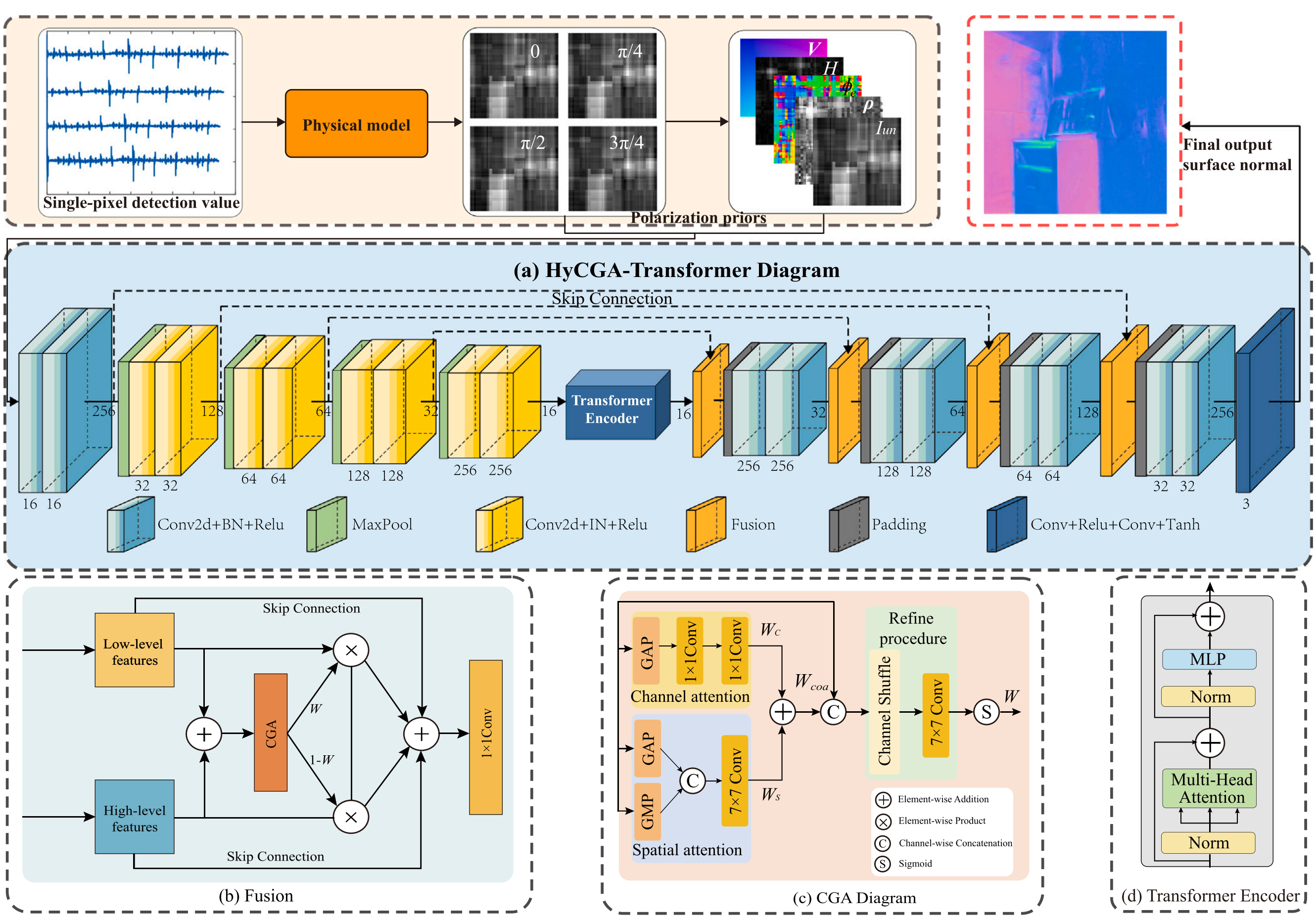


Fig. 2. Overall deep fusion network architecture.

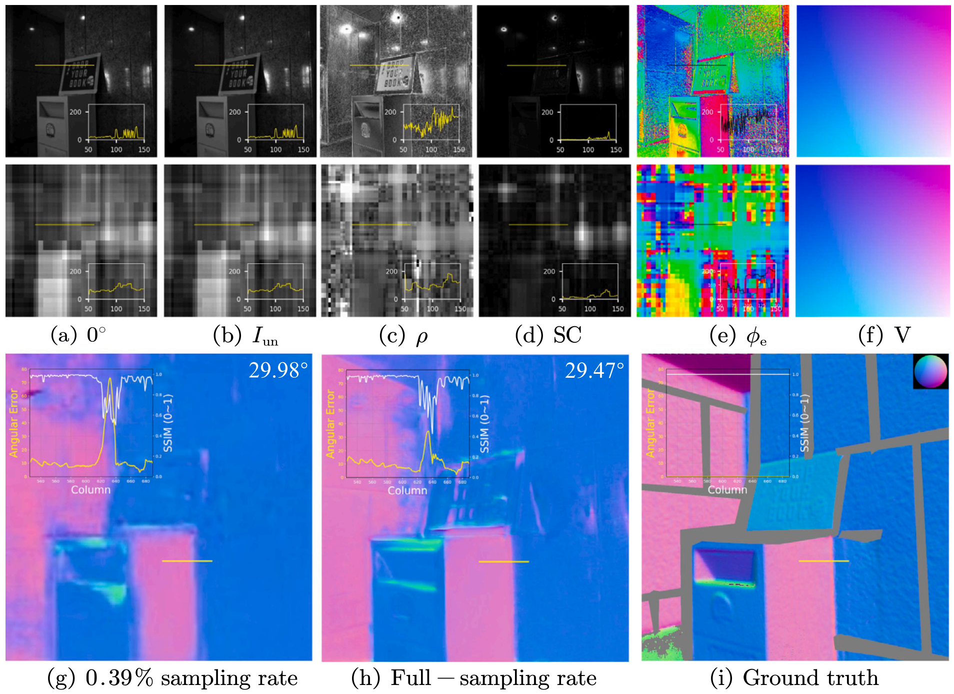


Fig. 3. Qualitative and quantitative comparison between undersampling and full-sampling results. The first row shows the full-sampling polarization-derived inputs, and the second row presents the corresponding results at a 0.39% sampling rate. (g) reconstructed surface normal at a 0.39% sampling rate. (h) corresponding surface normal under full-sampling. (i) Ground-truth surface normal. The curves at the bottom-right of the first and second rows show the pixel intensity profiles along the yellow sampling lines. The curves at the top-left of the third row show the angular error and structure Similarity Index Measure (SSIM) along the yellow sampling lines, with the mean angular error displayed at the top-right. A sphere is inserted in the ground truth to visually display the surface normal directions using color encoding.

128×128 and the number of measurements is $M = 64$, the sampling rate is $\mu = \frac{64}{128\times128} = 0.0039$.

Fig. 3 presents the qualitative and quantitative evaluation diagrams of undersampling and full-sampling results. The first row shows full-sampling input images, while the second row displays input images at a 0.39% sampling rate. The curves at the bottom-right of the first and second rows represent the pixel intensity profiles along the yellow sampling lines in the figures. As observed from the profiles, deep undersampling at 0.39% degrades the information of the original polarization images. Furthermore, significant aliasing artifacts of polarization information are evident in Fig. 3(e), increasing the difficulty of surface normal estimation for the algorithm. The angular error and Structural Similarity Index Measure (SSIM) curves at the top-left of Fig. 3(g) and Fig. 3(h) correspond to the yellow sampling lines, with the mean angular error displayed at the top-right. Analysis of these curves and the mean angular error demonstrates that comparable performance to full-sampling can still be achieved even under 0.39% deep undersampling.

2.2. Polarization prior information based on decoupled polarimetric dimensions

When the surface of an object composed of dielectric material is illuminated by completely unpolarized light, the reflected light becomes partially polarized. As shown in Fig. 4(b), a linear polarizer with a polarization angle ϕ_{pol} is placed in front of the camera. The intensity of each pixel in the captured image follows a sinusoidal variation under unpolarized illumination, as expressed by Eq. (3):

$$I(\phi_{\mathrm{pol}}) = I_{\mathrm{un}}[1 + \rho\cos(2(\phi_{\mathrm{pol}} - \phi))]. \tag{3}$$

Here, ϕ denotes the angle of polarization (AoP), ρ represents the degree of polarization (DoP), and I_{un} denotes the unpolarized intensity. The four captured polarization-state images are denoted by I_0, $I_{\pi/4}$, $I_{\pi/2}$, and $I_{3\pi/4}$. For visual clarity, only the intensity image corresponding to a polarization angle of zero is shown in Fig. 3(a). According to Fresnel theory, ϕ, ρ, and I_{un} are computed using Eqs. (4), (5), and (6), respectively.

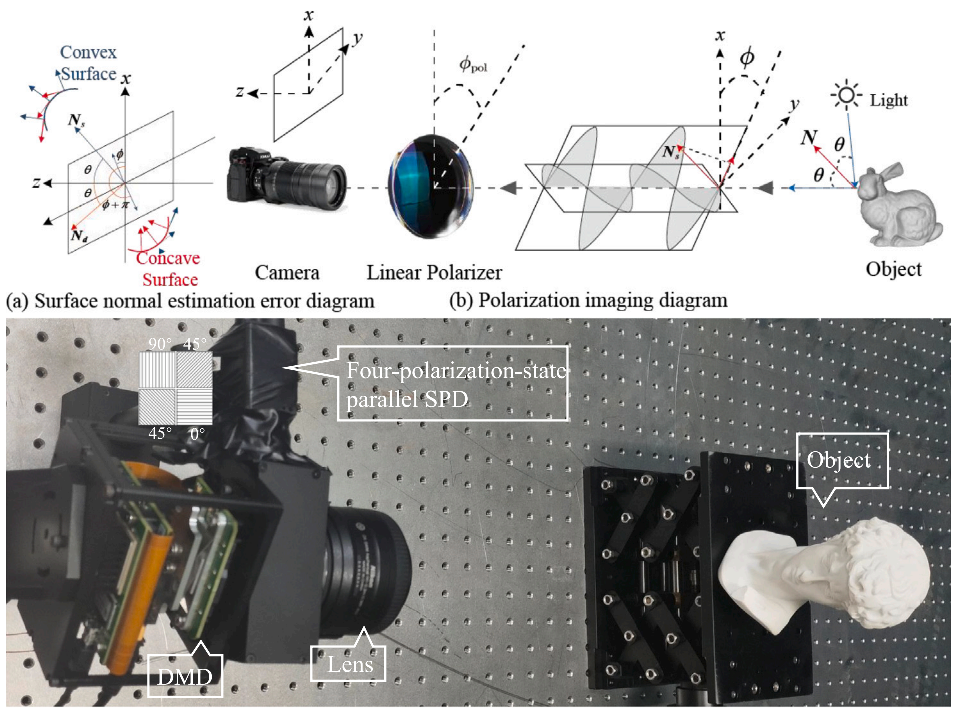
$$I_{\mathrm{un}} = (I_0 + I_{\pi/4} + I_{\pi/2} + I_{3\pi/4})/2, \tag{4}$$

$$\rho = \frac{\sqrt{(I_0 - I_{\pi/2})^2 + (I_{\pi/4} - I_{3\pi/4})^2}}{I_0 + I_{\pi/2}}, \tag{5}$$

$$\phi = \frac{1}{2}\arctan\left(\frac{I_{\pi/4} - I_{3\pi/4}}{I_0 - I_{\pi/2}}\right). \tag{6}$$

Among these, I_{un} (Fig. 3(b)) characterizes the overall two-dimensional texture information of the object surface, while ϕ and ρ (Fig. 3(c)) are related to the two angular components of the surface normal, respectively, thereby containing geometric information about the surface. Furthermore, by analyzing the $\cos(2(\phi_{pol}))$ term in Eq. (3), a π-radian ambiguity in the phase angle ϕ is identified: when two phase angles differ by π, the resulting intensity variation in the polarization image remains identical, leading to false concavity artifacts in the reconstructed object surface, as illustrated in Fig. 4(a). To overcome this issue, a polarization encoding strategy more efficient than existing phase angle representations was proposed by Lei et al. [14], in which the original phase angle ϕ is encoded to eliminate its drawbacks. Therefore, the phase angle is encoded using $\phi_e = (\cos 2\phi, \sin 2\phi)$, and the re-encoded phase angle ϕ_e is capable of more accurately describing the geometric characteristics of the surface (Fig. 3(e)).

In our SfP system decoupled by the polarization dimension, the normal surface N is represented by the azimuth angle φ and the zenith



(c) Single pixel imaging system with decoupling spatial and polarization dimensions.

Fig. 4. Surface normal imaging diagram. (a) the surface normal are defined by the azimuthal angle ϕ and zenith angle θ. Due to the π-radian ambiguity in ϕ, the convex surface (N_s) is incorrectly estimated as a concave surface (N_d). (b) Completely unpolarized light is partially polarized after reflecting off the object surface and is captured as a polarized image using a camera with a linear polarizer. (c) real-world experimental system.

angle θ, as shown in Eq. (7):

$$\boldsymbol{N} = \begin{bmatrix} N_x \\ N_y \\ N_z \end{bmatrix} = \begin{bmatrix} \cos\varphi\sin\theta \\ \sin\varphi\sin\theta \\ \cos\theta \end{bmatrix}. \tag{7}$$

When the diffuse component dominates in the reflection, the closed-form solution of θ could be computed by Eq. (8):

$$\cos\theta = \sqrt{\frac{\eta^4(1-\rho^2) + 2\eta^2(2\rho^2+\rho-1) + \rho^2 + 2\rho - 4\eta^3\rho\sqrt{1-\rho^2} + 1}{(\rho+1)^2(\eta^4+1) + 2\eta^2(3\rho^2+2\rho-1)}}, \tag{8}$$

where η denotes the refractive index. The azimuth angle is given by Eq. (9):

$$\varphi = \phi \text{ or } \phi + \pi. \tag{9}$$

It can be concluded from the multi-valued nature of the phase angle φ that the surface normal reconstruction process is prone to ambiguity.

In reconstruction tasks for complex scenes, large areas of specular reflection are often encountered, making it difficult to accurately estimate surface normals in these reflective regions. Therefore, drawing on previous research, specular confidence information is incorporated as prior knowledge into the system. The specular confidence (SC) (Fig. 3(d)) is calculated based on intensity variations among the four raw polarization images, as expressed by Eq. (10):

$$\mathrm{SC} = \min_{y \in I_{\mathrm{un}}} \left\{ \max\left\{I_0, I_{\pi/4}, I_{\pi/2}, I_{3\pi/4}\right\} - \min\left\{I_0, I_{\pi/4}, I_{\pi/2}, I_{3\pi/4}\right\} \right\}, \tag{10}$$

In our experiments, the patch size y is set to 3×3 by default. In addition, a viewing encoding V (Fig. 3(i)) is introduced to describe the viewing direction corresponding to each pixel during the imaging process [14].

In summary, the polarization prior information based on the decoupled polarimetric dimension is expressed as $\mathrm{Pol}_{\mathrm{prior}}$ = [images; I_{un}; ρ; ϕ_e; V; SC].

3. Network architecture

Benefiting from the global information capture capability of the Transformer [29] and the channel-spatial fusion capacity of the content-guided attention (CGA) mechanism [30], we propose a deep fusion network named HyCGA-Transformer, which integrates a hybrid fusion strategy based on CGA and a Transformer Encoder. As shown in Fig. 2(a), ultra-low-sampled polarization detection values of the target scene are acquired using a four-polarization-state parallel single-pixel detector (SPD). An undersampled polarization image is then constructed via the SPI system described in Section 2.1, yielding undersampled polarization information for the four-polarization states. Subsequently, encoded polarization information is obtained using the polarization prior introduced in Section 2.2. Finally, the undersampled polarization images of the four-polarization states and the five types of physical prior information are jointly fed into the network to recover high-precision surface normal information.

Overall Network Architecture: The objective of this work is to achieve accurate surface normal estimation through an SPI ultra-low-sampling system with decoupled spatial and polarimetric dimensions. To this end, a deep fusion network named HyCGA-Transformer is proposed. As shown in Fig. 2, the overall network consists of four downsampling modules, four upsampling modules, one Transformer Encoder module, four hybrid fusion modules based on CGA, and one output MLP module. The downsampling and upsampling modules encode and decode the polarization information, respectively. The Transformer Encoder module (detailed in Fig. 2(d)) captures global dependencies within the polarization information through multi-head self-attention mechanisms. The CGA-based hybrid fusion modules (detailed in Fig. 2(b)) deeply integrate polarization prior information via four-level skip connections.

By learning spatial and channel-wise attention weights, these modules adaptively modulate features, enhancing their representational capacity. Finally, the MLP module adjusts the channel dimensions and image size to correctly output the surface normal map.

Hybrid Fusion Mechanism Based on CGA: To improve feature extraction efficiency, we introduce a hybrid fusion scheme based on CGA, whose core idea is to adaptively modulate features via learned spatial and channel weights. As shown in Fig. 2(b), the details of the CGA-based hybrid fusion scheme [30] are illustrated. The key component is the use of CGA to enhance feature representation. Traditional feature attention modules suffer from limited feature extraction capability in complex scenes due to the lack of interaction between channel and spatial attention, and the fact that spatial attention only captures image-level information distribution. To address this, we adopt a coarse-to-fine CGA mechanism that generates unique spatial importance maps for each channel, enabling dynamic interaction between channel and spatial dimensions. The implementation details of CGA are shown in Fig. 2(c). The channel attention weights and spatial attention weights are generated by CGA through Eqs. (11) and (12), respectively.

$$W_c = C_{1\times1}(\max\left(0, C_{1\times1}(X^c_{GAP})\right)), \tag{11}$$

$$W_s = C_{7\times7}([X^s_{GAP}, X^s_{GMP}]), \tag{12}$$

where $\max(0,x)$ denotes the ReLU activation function, $C_{k\times k}(\cdot)$ represents a convolutional kernel with kernel size $k\times k$, and $(\cdot)$ indicates the channel-wise concatenation operation. X^c_{GAP}, X^s_{GAP}, and X^s_{GMP} correspond to features processed by global average pooling across the channel dimension, global average pooling across the spatial dimension, and global max pooling across the spatial dimension, respectively. We then fuse W_c and W_s through a simple addition operation following broadcasting rules to obtain the coarse Spatial Importance Map (SIM), as shown in Eq. (13):

$$W_{coa} = W_c + W_s, \tag{13}$$

where W_{coa} and X are aligned channel-wise since W_c is channel-based. To obtain the final refined SIM W, each channel of W_{coa} is further adjusted based on the corresponding input features. We utilize the content of the input features to guide the generation of the final channel-specific SIM. Specifically, as shown in Eq. (14), W_{coa} and the channels of X are interleaved and rearranged through a channel shuffle operation $CS(\cdot)$.

$$W = \sigma(\mathcal{G}_{7\times7}(CS([X, W_{coa}]))), \tag{14}$$

where σ denotes the sigmoid operation, $\mathcal{G}_{k\times k}$ indicates a group convolution layer with a kernel size of $k\times k$. The computation process of the final features is given by Eq. (15):

$$F_{fuse} = C_{1\times1}(F_{low}\cdot W + F_{high}\cdot(1 - W) + F_{low} + F_{high}), \tag{15}$$

where 1 denotes a matrix of all ones, used to perform the corresponding addition operation, and $C_{1\times1}$ represents a 1×1 convolutional layer.

Transformer Encoder: To capture global features from the encoder feature maps, we introduce a Transformer Encoder at the intermediate stage of the encoder–decoder architecture (as shown in Fig. 2(d)) to leverage global contextual information. We omit the linear projection layers traditionally used in Transformers; furthermore, since the viewing encoding V provides information analogous to positional embedding, no additional positional embedding is introduced [14].

Loss function: To supervise normal estimation, we employ the cosine similarity loss to penalize the cosine distance between the ground truth surface normal and the predicted normal [12]. The loss function for supervising the SfP imaging architecture is given by Eq. (16):

$$\mathcal{L}_{cos} = \frac{1}{W\times H}\sum_{i}^{W}\sum_{j}^{H}(1 - \langle \hat{\boldsymbol{N}}_{ij}, \boldsymbol{N}_{ij}\rangle), \tag{16}$$

where $\langle\cdot,\cdot\rangle$ denotes the dot product, $\hat{\boldsymbol{N}}_{ij}$ represents the estimated surface normal at pixel position (i,j), and $\boldsymbol{N}_{ij}$ is the ground truth surface normal corresponding to the surface. By minimizing the loss, we can effectively enhance the model's accuracy in reconstructing surface normals.

4. Experimental results and discussion

Network Training Details: The network model in this study was constructed and trained based on the PyTorch framework. Training was performed on a single NVIDIA A800 GPU with a batch size of 16 for a maximum of 10k iterations, and the overall training process took approximately 3 days. To balance training speed and convergence performance, the Adam optimizer was used with a learning rate set to $1e-4$.

To improve model generalization and robustness, input images from the two public datasets, DeepSfP and SPW, were cropped into 256×256 patches. SPI imaging was then simulated using a 65,536-order Hadamard matrix with cake-cutting ordering, where the light-field size was 256×256. For the comparative experiments, four sampling rates, namely 0.39%, 1.56%, 6.25%, and 100%, were adopted. In addition, to analyze the effect of the sampling rate on imaging quality, eight sampling rates, i.e., 0.39%, 1.56%, 6.25%, 12.5%, 25%, 50%, 75%, and 100%, were considered.

For the real-world dataset, SPI imaging was performed using a 16,384-order Hadamard matrix with cake-cutting ordering, with a light-field size of 128×128. Three sampling rates, namely 1.56%, 6.25%, and 100%, were used in the comparative experiments. For the analysis of the effect of sampling rate on imaging quality, seven sampling rates, i.e., 1.56%, 6.25%, 12.5%, 25%, 50%, 75%, and 100%, were further considered.

Evaluation metrics: To comprehensively evaluate the performance of the proposed method in the estimation of surface normals, this study follows previous related works [6,14] and adopts six metrics to quantitatively analyze the precision of the predicted normals. These include three angular error metrics—mean angular error, median angular error, and root mean square angular error (lower values indicate better performance)—and three accuracy proportion metrics—the proportion of pixels with angular errors below 11.25°, 22.5°, and 30° (higher values indicate better performance).

The first three angular error metrics are primarily used to quantify the angular deviation between the predicted normal and the ground truth, providing detailed error analysis. The latter three metrics measure the proportion of pixels whose predicted normals fall within the specified angular thresholds (11.25°, 22.5°, and 30°), reflecting the model's performance under varying accuracy requirements. Through this comprehensive evaluation framework, this study thoroughly demonstrates the precision and reliability of the proposed method in surface normal estimation across diverse scenarios and conditions.

4.1. Datasets

In this study, two typical datasets from publicly available SfP datasets, namely DeepSfP [12] and SPW [14], were selected for experimental simulation and validation. In addition, a real-world scene dataset was collected through the designed imaging system to verify the applicability of the system under practical conditions.

DeepSfP: DeepSfP is the first publicly available object-level SfP dataset. Although each image contains only one object, the surface normals are of high quality. The model was trained using the same training/testing split as provided in the original DeepSfP paper [12].

SPW: SPW is the first publicly available scene-level SfP dataset, consisting of 522 sets of images from 110 different scenes. Each scene contains multiple objects composed of different materials with varying reflective properties, and multiple highly reflective areas may appear within a single scene. In the original paper [14], 403 sets were used for training and 119 for testing. However, only 79 sets are available for testing in the publicly released version of the SPW dataset. Therefore, 403 sets were used for training and 79 sets for testing.

Real-World Dataset: The principles and equipment details are illustrated in Fig. 1. A camera lens (focal length 420–800 mm, aperture

F3.8-16) images target information onto the light modulation surface of a DMD (Vialux DLP650LNIR). The DMD loads a 16,384-order Hadamard matrix (light field size 128×128) based on the cake-cutting sequence to modulate the target. Due to the DMD's $\pm 12^{\circ}$ working mechanism, additional mirrors are placed along the $\pm 12^{\circ}$ propagation directions to alter the transmission path, which is then focused by a lens onto a four-polarization-state parallel single-pixel detector (SPD). The detector acquires modulated signals of four polarization states, and finally, polarization images of the four states are obtained via the SPI principle.

Comparative experiments were conducted on the DeepSfP and SPW datasets with three physics-based methods (Miyazaki et al. [31], Mahmoud et al. [4], Smith et al. [5]) and three approaches based on deep learning (Ba et al. [12], Lei et al. [14], Lyu et al. [32]). For the model by Ba et al. [12], due to the absence of evaluation metrics in the original paper, it was retrained on both datasets to obtain the corresponding metrics. For the method proposed by Lei et al. [14], a mismatch was observed between the number of training/test sets described in the paper and the actually available images in the published SPW dataset. Therefore, the model was retrained on both datasets using the publicly released source code and the parameters provided in the paper. For the approach introduced by Lyu et al. [32], since no visual results on the DeepSfP and SPW datasets were published in the article, the model was retrained on both datasets to acquire both visual results and evaluation metrics.

4.2. Evaluation on DeepSfP dataset

A comparative experiment was first conducted on the DeepSfP dataset. The visual results of five objects with surface textures ranging from simple to complex are presented in Fig. 5. Among them, the Christmas ornament and vase, which have relatively simple surface textures, were used to demonstrate the differences in the ability of various methods to reconstruct smooth surface normals. As clearly shown in Fig. 5, larger angular errors were observed in the reconstruction results of physics-based methods. In contrast, the three deep learning-based methods achieved better surface normal reconstruction results.

For our SPI ultra-low-sampling SfP method, comparisons were made at four sampling rates: 0.39%, 1.56%, 6.25%, and 100%. As shown in Fig. 5, among the deep learning-based methods, the approaches by Ba et al. [12] and Lei et al. [14] were able to reconstruct relatively clear surface normals, but some artificial textures were also introduced.

Table 1
Quantitative evaluation on the DeepSfP dataset, where the best-performing results are highlighted in bold black font (↓ denotes lower is better, ↑ denotes higher is better).

Method	Angular Error ↓			Accuracy ↑		
	Mean	Median	RMSE	11.25°	22.5°	30.0°
Miyazaki et al.	45.38	43.69	49.39	3.40	13.05	23.64
Mahmoud et al.	47.26	44.13	52.47	3.98	15.01	26.41
Smith et al.	50.49	49.78	54.60	2.99	11.28	18.51
Ba et al.	17.72	14.59	21.99	37.68	72.45	85.59
Lei et al.	15.47	11.74	20.01	48.12	78.39	88.22
Lyu et al.	15.07	11.29	19.55	49.83	79.27	88.67
Ours-0.39% sampling	15.91	12.33	20.46	45.54	77.99	87.82
Ours-1.56% sampling	15.81	12.12	20.58	46.41	78.09	88.03
Ours-6.25% sampling	14.95	11.13	19.40	50.46	79.32	88.70
Ours full-sampling	**13.50**	**9.91**	**17.82**	**55.95**	**82.93**	**90.68**

In comparison, with our method, the results at a 6.25% sampling rate maintained smooth surface reconstruction while accurately preserving details. For the results at a 1.56% sampling rate, although some detailed information was lost, the overall error remained small. For the results at a 0.39% sampling rate, relatively high-quality reconstructions were still obtained for objects with simple textures, such as the Christmas ornament and vase, but the reconstruction quality was poor for objects with complex surface textures (dragon, flamingo, and horse), failing to adequately reconstruct the surface normals.

Quantitative results on the DeepSfP dataset are summarized in Table 1. As expected, deep learning-based methods significantly outperform traditional physics-based methods. For the proposed undersampled method, the error gradually decreases as the sampling rate increases. When the sampling rate is reduced to 0.39%, all six quantitative metrics are inferior to those of the methods proposed by Lei et al. [14] and Lyu et al. [32]. At this extremely low sampling rate, substantial surface details are lost, and the network cannot fully recover fine geometric structures, thereby limiting the overall reconstruction accuracy. When the sampling rate increases to 6.25%, more local details are preserved, and the proposed method achieves angular error and accuracy results comparable to those of the other three deep learning-based methods. Under full-sampling (100%), the proposed method achieves the best overall performance in both angular error and accuracy.

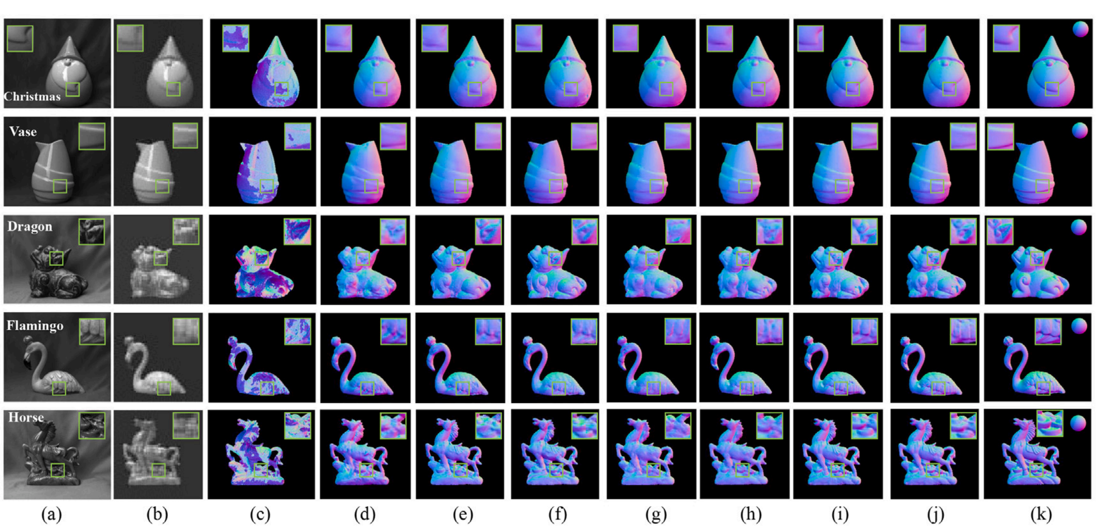


Fig. 5. Visual comparison results on the DeepSfP dataset. (a) Five objects with shapes ranging from simple to complex. (b) imaging results at a 0.39% sampling rate. (c) Miyazaki et al. (d) Ba et al. (e) Lei et al. (f) Lyu et al. (g) Ours-0.39% sampling rate. (h) Ours-1.56% sampling rate. (i) Ours-6.25% sampling rate. (j) Ours full-sampling. (k) the ground truth. The large boxes represent magnified views of the corresponding small boxes. A sphere is inserted in the ground truth to visually display the surface normal directions using color encoding.

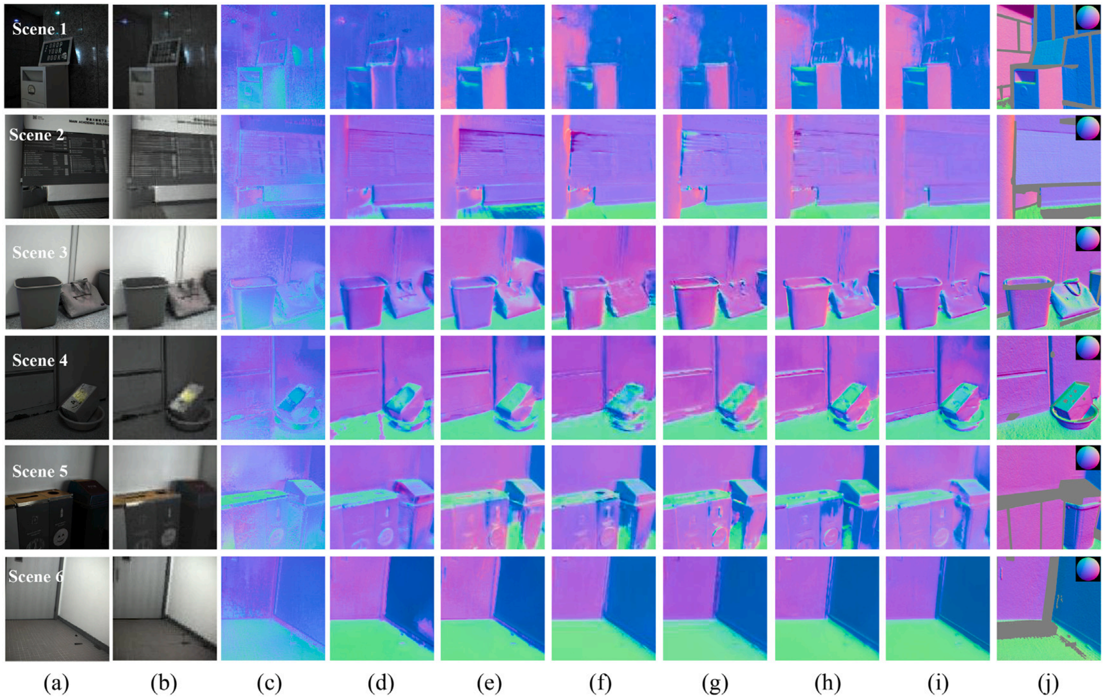


Fig. 6. Visual comparison results on the SPW dataset. (a) Fully sampled images of six representative scenes. (b) Imaging results at a 0.39% sampling rate. (c) Ba et al. (d) Lei et al. (e) Lyu et al. (f) Ours-0.39% sampling rate. (g) Ours-1.56% sampling rate. (h) Ours-6.25% sampling rate. (i) Ours full-sampling. (j) The ground truth. A sphere is inserted into the ground truth to visually display the surface normal directions using color encoding.

4.3. Evaluation on SPW dataset

A comparative experiment was conducted on the SPW dataset, and six representative scenes were selected for visual presentation. Each scene includes objects with at least three different material properties (i.e., varying reflectivity and albedo). For example, Scene 1 contains three distinct materials: a tile wall, a metal box, and a blackboard. For our SPI ultra-low-sampling SfP method, the results were compared at four sampling rates, namely 0.39%, 1.56%, 6.25%, and 100%. The visual results are shown in Fig. 6. For the SPW dataset, which comprises environments with multiple discontinuous surfaces and high scene complexity, the method by Ba et al. [12] struggled to handle discontinuous surfaces, resulting in reconstructed outputs with holes and loss of textural details. The methods by Lei et al. [14] and Lyu et al. [32] were prone to normal errors in specular reflection regions (e.g., the wall in Scene 1). In contrast, the proposed method is able to recover usable surface normal results across all four sampling rates, and its reconstruction quality improves consistently as the sampling rate increases, even for scenes with diverse material properties and both diffuse and specular reflections.

It is noteworthy that, since the scene-level SPW dataset does not require as fine-grained surface details as DeepSfP, the visual results of surface normals at a sampling rate of 0.39% already achieve better performance compared to the other three methods. Quantitative results of all methods on the SPW dataset are provided in Table 2. A clear advantage of deep learning-based methods over traditional physics-based modeling approaches can be observed. At the sampling rate of 0.39%, all six quantitative metrics of the proposed method outperform those of Ba et al. [12], while only two metrics exceed those of Lei et al. [14], and all six metrics remain inferior to the method developed by Lyu et al. [32]. This gap can be attributed to the high complexity of the SPW dataset, which contains coupled illumination effects and diverse material properties, including highly reflective regions in Scene 1 of Fig. 6 and challenging mixed-material conditions in Scene 3, where plastic, fabric, wood, ceramic, and other materials coexist. Under the extremely low sampling rate of 0.39%, the valid observation information becomes severely insufficient, leading to noticeable errors in all compared methods. When the sampling rate increases to 1.56%, all methods show clear performance improvement compared with the 0.39% setting. After the sampling rate reaches 6.25%, the proposed method achieves better angular error and accuracy results in multiple metrics than Lyu et al. [32]. Under full-sampling (100%) conditions, the proposed method achieves the best overall quantitative performance among all competing approaches.

Table 2
Quantitative evaluation on the SPW dataset, where the best-performing results are highlighted in bold black font (↓ denotes lower is better, ↑ denotes higher is better).

Method	Angular Error ↓			Accuracy ↑		
	Mean	Median	RMSE	11.25°	22.5°	30.0°
Miyazaki et al.	54.01	51.60	59.69	3.97	13.87	22.37
Mahmoud et al.	53.13	51.93	57.02	2.98	12.37	21.31
Smith et al.	49.35	47.26	51.87	11.33	24.86	34.54
Ba et al.	36.96	33.53	42.66	11.26	33.52	47.42
Lei et al.	23.79	17.23	32.78	43.01	70.47	78.93
Lyu et al.	22.72	16.26	31.81	46.34	73.04	80.38
Ours-0.39% sampling	22.78	17.69	29.38	32.55	64.66	76.66
Ours-1.56% sampling	20.96	15.55	27.92	39.96	70.95	79.96
Ours-6.25% sampling	19.66	14.62	26.43	42.79	73.03	81.74
Ours full-sampling	**17.27**	**12.35**	**23.82**	**50.11**	**78.95**	**85.69**

4.4. Evaluation on real-world dataset

The constructed SPI ultra-low-sampling SfP system based on decoupled spatial and polarimetric dimensions was evaluated in real-world scenarios to comprehensively assess its performance. Experiments were conducted using light field information at three sampling rates (1.56%, 6.25%, and 100%), corresponding to 256, 1024, and 16,384 light field modulations implemented with a cake-cutting-ordered 16,384 order Hadamard matrix (optical field size 128 × 128). It should be noted

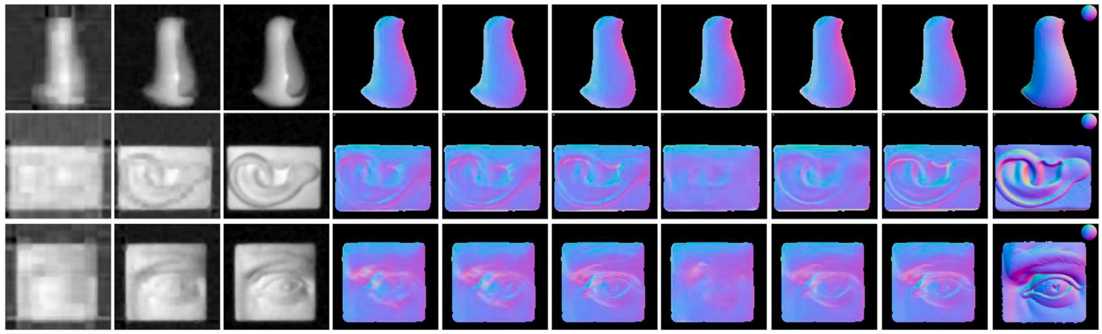


Fig. 7. Visual results on the real-world dataset. (a) imaging results at a 1.56% sampling rate. (b) imaging results at a 6.25% sampling rate. (c) imaging results at a 100% sampling rate. (d) Ba et al. (e) Lei et al. (f) Lyu et al. (g) Ours-1.56% sampling rate. (h) Ours-6.25% sampling rate. (i) Ours full-sampling. (j) the ground truth. A sphere is inserted in the ground truth to visually display the surface normal directions using color encoding.

Table 3
Quantitative evaluation on the real-world dataset, where the best-performing results are highlighted in bold black font (↓ denotes lower is better, ↑ denotes higher is better).

Method	Angular Error ↓			Accuracy ↑		
	Mean	Median	RMSE	11.25°	22.5°	30.0°
Ba et al.	6.97	2.12	12.30	76.92	90.24	94.55
Lei et al.	6.74	3.88	11.28	80.33	91.88	95.42
Lyu et al.	6.20	2.28	11.06	81.23	91.93	95.46
Ours-1.56% sampling	9.98	4.88	15.98	63.03	83.11	91.01
Ours-6.25% sampling	5.41	1.67	10.25	85.31	92.58	95.75
Ours full-sampling	**5.26**	**1.38**	**9.99**	**86.34**	**92.76**	**95.98**

that during real-world data acquisition, imperfect coupling between the DMD chip and the imaging lens may introduce lens distortion, thereby generating noisy signals.

Visual results from the real-world experiments are presented in Fig. 7. It can be observed that for objects with simple surface texture details (row 1), a low sampling rate is sufficient to achieve surface normal reconstruction results consistent with full-sampling. In contrast, for objects rich in surface details (rows 2 and 3), the detail recovery capability of surface normals is significantly enhanced as the sampling rate increases. These results validate the effectiveness of our undersampled system in performing SfP imaging on objects with relatively simple surface textures.

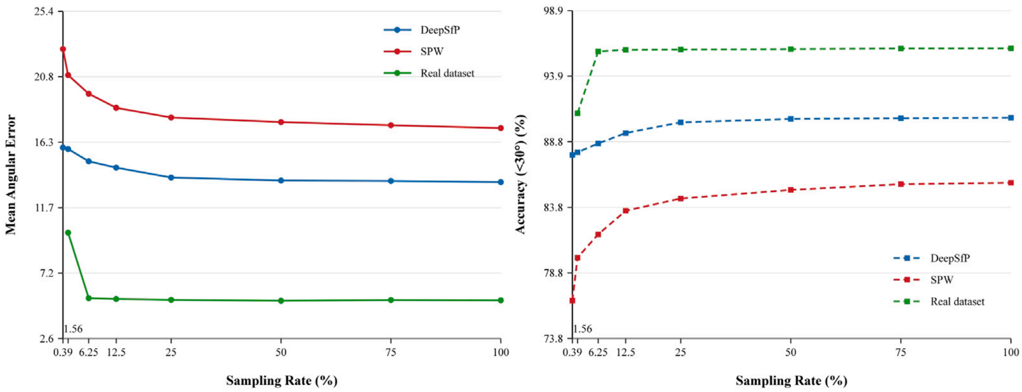
Table 3 presents the quantitative comparison between the proposed system and three deep learning-based SfP methods under three sampling rates on the real-world dataset. The results show that, due to system errors and other non-ideal factors in real-world experiments, severe loss of object texture and contour details occurs at the 1.56% sampling rate, causing all six evaluation metrics of the proposed method to fall below those of the three competing methods. When the sampling rate is increased to 6.25%, both the visual quality and the quantitative performance of the proposed method surpass those of all three comparison methods. Fig. 8 further illustrates how reconstruction quality varies with the sampling rate. Overall, these results indicate that, in SfP imaging, a substantial increase in sampling rate does not lead to a proportional improvement in surface normal reconstruction quality. Therefore, using fully sampled images as input is not always necessary, which further highlights the feasibility and importance of reducing the sampling rate.

It should be noted that SPI requires multiple coding patterns to be sequentially projected while the target remains in the same state [33]. Target motion can therefore break the consistency among different measurements [34], and abrupt changes in ambient illumination can alter the baseline intensity of the bucket signal. For this reason, the current system is not intended for non-ideal dynamic conditions; instead, it mainly validates the imaging capability for static targets. In future work, robustness to object motion and illumination fluctuations could be improved by increasing the DMD refresh rate, shortening the sampling sequence, introducing synchronized exposure control, and incorporating reference-channel normalization.

4.5. Data and computational efficiency analysis

To verify the advantages of the proposed method in data acquisition, transmission, and backend reconstruction efficiency, comparative experiments between the traditional full-sampling SfP method and the proposed SPI ultra-low-sampling SfP method are carried out under the same input resolution and hardware conditions. Conventional SfP typically requires capturing full-resolution images at four polarization angles, i.e., I_0, I_{45}, I_{90}, and I_{135}. As shown in Table 4, taking single-frame data with a resolution of 1024×1024 and 8-bit grayscale storage per pixel as an example, the traditional method needs to store and transmit $4 \times 1024 \times 1024$ pixel values per frame, corresponding to a data volume of 4.00 MiB. In contrast, the proposed method obtains bucket detection measurements of four polarization channels simultaneously through a four-quadrant parallel single-pixel detector. At sampling rates of 0.39%, 1.56%, and 6.25%, only 256, 1024, and 4096 measurements are required for each 256×256 sub-block, respectively. Since a 1024×1024 image is divided into 16 256×256 sub-blocks, the single-frame raw data volumes of the proposed method at the above three sampling rates are 0.0156 MiB, 0.0625 MiB, and 0.25 MiB, corresponding to data volume reductions of 256, 64, and 16 times compared with the traditional method.

Transmission bandwidth is approximately linearly correlated with the raw data volume per frame. For quantitative comparison, this paper adopts a common video frame rate of 30 fps as the benchmark for equivalent bandwidth conversion, where the equivalent transmission bandwidth is calculated as the product of the raw per-frame data volume and 30. It should be emphasized that the 30 fps is only used to compare the bandwidth requirements of different sampling strategies under the assumption of a unified frame rate, rather than representing the actual end-to-end imaging frame rate of the current prototype system. At the same frame rate, the conventional full-sampling SfP method needs to transmit complete four-polarization images, whereas the proposed method only transmits compressed measurement sequences of four polarization channels. Neglecting the overhead of communication protocols and data encapsulation, the bandwidth demand of the proposed scheme decreases proportionally with the raw per-frame data volume. Taking the sampling rate of 0.39% as an example, its theoretical transmission bandwidth is only 1/256 of that of the traditional four-polarization image acquisition scheme. Experimental results verify that the SPI undersampling mechanism effectively reduces storage and



(a) Comparison of mean angular error (lower is better) at different sampling rates on the three datasets.

(b) Comparison of accuracy (<30°) (higher is better) at different sampling rates on the three datasets.

Fig. 8. Comparison of the relationship between different sampling rates and reconstruction quality across the three datasets, including two simulated datasets (DeepSfP and SPW) and a real-world dataset, measured by mean angular error and accuracy within 30°.

Table 4
Comparison of data storage, equivalent transmission bandwidth, model parameter, forward inference time and end-to-end time. The input resolution is 1024×1024, the bit depth is 8-bit grayscale, and the equivalent bandwidth is calculated at 30 fps.

Method	Raw data (per frame)	Storage (MiB)	Bandwidth (MiB/s; reduction)	Params (M)	Inference (ms)	End-to-end (ms)
Ba et al.	$4 \times 1024 \times 1024$	4.0000	120.0000; 1×	10.93	51.99	52.01
Lei et al.	$4 \times 1024 \times 1024$	4.0000	120.0000; 1×	42.48	122.31	122.44
Lyu et al.	$4 \times 1024 \times 1024$	4.0000	120.0000; 1×	166.12	7419.87	7429.02
Ours-0.39% sampling	$4 \times 16 \times 256$	0.0156	0.4688; 256×	33.04	976.53	981.96
Ours-1.56% sampling	$4 \times 16 \times 1024$	0.0625	1.8750; 64×	33.04	976.53	988.34
Ours-6.25% sampling	$4 \times 16 \times 4096$	0.2500	7.5000; 16×	33.04	976.53	1021.41

transmission costs at the data acquisition source, showing strong potential for static 3D imaging tasks with limited bandwidth and storage resources.

It should be noted that the reduction in data volume does not mean that the end-to-end reconstruction time is reduced by the same proportion. For traditional deep learning-based SfP methods, the backend computation mainly includes polarization feature preprocessing and network inference. For the proposed method, in addition to polarization feature calculation and network inference, it also involves the SPI physical decoding process to recover the undersampled polarization images from the bucket detection measurements. Therefore, to objectively evaluate the computational efficiency, the forward inference time and end-to-end reconstruction time of the model are counted separately in this paper. The model forward inference time only includes the process from data input into the network to surface normal output. The end-to-end reconstruction time includes SPI decoding, polarization feature calculation, and network forward propagation. All methods are tested on the same NVIDIA A800 GPU with the same batch size and input resolution.

Table 4 presents the comparison results between the traditional SfP method and the proposed method in terms of storage occupation, equivalent transmission bandwidth reduction factor, model parameters, and reconstruction time. It can be seen that the traditional full-sampling SfP method requires the transmission of complete four-polarization images, resulting in high raw data volume and bandwidth requirements. In contrast, the proposed method can significantly reduce the data volume and transmission burden at the acquisition end under a 0.39% sampling rate, and achieves a good balance between low data overhead and high reconstruction accuracy at 1.56% and 6.25% sampling rates. In terms of computational time consumption, the end-to-end time of the proposed method is jointly affected by SPI decoding and network inference. Therefore, the main advantages of the current system are reflected in low-data-volume acquisition and low-bandwidth transmission. In the future, the reconstruction time can be further reduced by adopting higher refresh rate DMDs, high-speed detectors, GPU/FPGA parallel SPI decoding, and lightweight networks that directly recover surface normals from compressed measurements.

4.6. Analysis of sampling rate and reconstruction quality

To systematically evaluate the effect of sampling rate on reconstruction quality, comparative experiments were conducted on the DeepSfP and SPW datasets using eight sampling rates, namely 0.39%, 1.56%, 6.25%, 12.5%, 25%, 50%, 75%, and 100%. For the real-world dataset, however, only seven sampling rates, i.e., 1.56%, 6.25%, 12.5%, 25%, 50%, 75%, and 100%, were considered. This is because, during real-world data acquisition, factors such as imperfect lens-DMD coupling, non-uniform illumination distribution, and strong system error interference lead to severe loss of target texture and contour details at the extremely low sampling rate of 0.39%. The corresponding relationship between sampling rate and reconstruction quality is shown in Fig. 8. As the sampling rate increases, the mean angular error (lower is better) on all three datasets exhibits a decreasing trend, as shown in Fig. 8(a), while the proportion of pixels with angular error below 30° (higher is better) shows an increasing trend, as shown in Fig. 8(b). The most significant performance improvement is observed in the low-sampling-rate range of 0.39%–6.25%. When the sampling rate further increases beyond 25%, the performance gain gradually saturates, indicating that although higher sampling rates can still improve reconstruction quality, the marginal benefit becomes weaker.

Representative visual results are presented in Fig. 9. The experimental results demonstrate that at a 1.56% sampling rate, the system still exhibits limited reconstruction capability for regions with rich textural details (such as the feathers of the flamingo in the second row and the hair of the David statue in the third row). In contrast, for regions with relatively simple textures (such as the object in the first row and the neck of the flamingo in the second row), a 1.56% sampling rate achieves visual results nearly consistent with those of full-sampling. When the

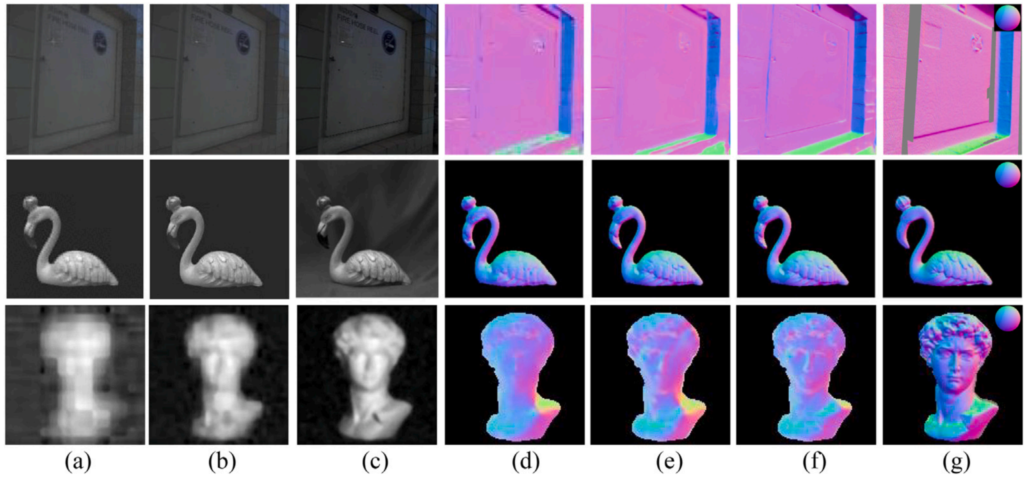


Fig. 9. Importance of undersampling. (a) SPI at 1.56% sampling rate; (b) SPI at 6.25% sampling rate; (c) fully sampled image; (d) surface normal map at 1.56% sampling rate; (e) surface normal map at 6.25% sampling rate; (f) surface normal map at full-sampling rate; (g) the ground truth. A sphere is inserted into the ground truth to visually display the surface normal directions using color encoding.

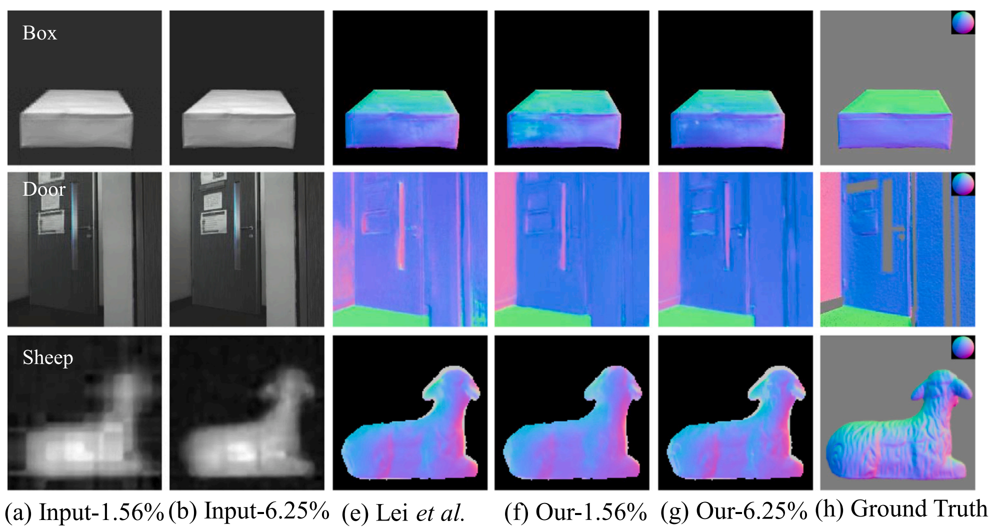


Fig. 10. Qualitative comparison of reconstruction results for representative objects with different material properties, including plastic, wood, sculpture, and ceramic, under varying input sampling rates.

sampling rate is increased to 6.25%, the reconstruction results become highly similar visually to those under full-sampling. These results clearly demonstrate that the undersampling technique can substantially reduce data volume while still maintaining high imaging quality, thereby highlighting its important value in improving system efficiency.

From the perspective of balancing sampling rate, computational cost, and reconstruction accuracy, a lower sampling rate effectively reduces the number of DMD modulations, detector readouts, and transmitted data, but it also aggravates undersampling-induced aliasing and increases the difficulty of surface normal recovery. By contrast, a higher sampling rate provides more complete spatial and polarimetric information, although the acquisition time and data volume increase approximately linearly with the number of measurements. Therefore, a sampling rate of 0.39% is more suitable for low-data applications involving objects with simple textures; 1.56% provides a practical trade-off between data cost and reconstruction accuracy; and 6.25% is more appropriate for targets with multiple materials, strong reflectivity, and rich texture details, where more robust and accurate 3D reconstruction is required.

4.7. Material attribute analysis

Due to the distinct reflective properties of polarized light on dielectric and non-dielectric materials, a comparative experiment was designed to evaluate the model's generalization capability across various material types, based on the models with sampling rates of 1.56% and 6.25%. As shown in Fig. 10, the three selected samples include multiple materials such as plastic, wood, sculpture, and ceramic, covering both diffuse and specular reflection types. Qualitative results demonstrate that the model effectively handles both diffuse and specular reflective materials, exhibiting strong reconstruction performance on surfaces of both dielectric and non-dielectric materials.

It is worth noting that real-world data acquisition is affected by several non-ideal factors, including imperfect coupling between the lens and the DMD, non-uniform illumination distribution, and noticeable system-induced errors. At a sampling rate of 1.56%, severe loss of reconstruction details can already be observed in some targets, such as the Sheep in the third row of Fig. 10(a). Therefore, we did not include the extremely low 0.39% sampling setting in the real-world experiments. Nevertheless, the multi-sampling-rate comparisons on the DeepSfP and SPW simulation datasets, where the 0.39% sampling rate is included, further indicate that the material generalization capability of the model is highly correlated with the sampling rate. At the extremely low sampling rate of 0.39%, the model can only recover the main surface normal structure of objects with simple materials. For complex materials with high reflectivity, strong textures, or rich details, it is prone to loss of normal details and degradation in reconstruction accuracy. When the sampling rate is increased to 1.56% and 6.25%, the reconstruction stability and fidelity of material boundaries, specular reflection regions, and local texture features are substantially improved.

In the above experiments, extensive validation was carried out in indoor single-object scenarios as well as outdoor static scenes containing multiple objects with different material properties. The visual results show that, even under low sampling rates, long imaging distances, and complex backgrounds, the proposed SPI ultra-low-sampling SfP system with decoupled spatial and polarimetric dimensions is still capable of reconstructing high-quality surface normals. These findings demonstrate the strong potential of combining SPI with SfP, not only for high-throughput post-processing applications, but also for the development of image-free intelligent sensing technologies and future innovative sensing systems.

In summary, through extensive testing in target scenes, including cases with high reflectivity, multiple material properties, and textureless surfaces, as well as in static scenes, the effectiveness and robustness of the SPI ultra-low-sampling SfP approach based on decoupled spatial and polarimetric dimensions have been validated. The combination of SPI and SfP sensing holds both theoretical and practical significance. It lays a foundation for practical applications and is expected to play a key role in the development of innovative sensing technologies in the future.

4.8. Ablation studies and analyses

Ablation studies were conducted at a sampling rate of 6.25% on two simulated datasets and one real-world scene dataset to analyze the importance of each design component in our architecture. The quantitative evaluation results are summarized in Table 5, and the visual comparison results are presented in Fig. 11. An error analysis was designed for each component of our framework, and the impact of removing each part can be clearly observed through the visual results.

First, the effectiveness of each component in the polarization priors was validated. Five controlled experiments were conducted: (i) removal of I_{un}; (ii) removal of ϕ_e; (iii) removal of ρ; (iv) removal of V; and (v) removal of SC. The quantitative comparison results are presented in Table 5. It can be observed that removing any component of the polarization priors leads to degraded fusion performance, demonstrating the rationale of the polarization prior construction.

To validate the effectiveness of our proposed deep fusion module, three controlled experiments were conducted: (i) the CGA module was removed and replaced with simple additive operations; (ii) the CGA-based hybrid fusion module was removed and replaced with basic skip connections; and (iii) the Transformer Encoder module was removed, with only skip connections used between the downsampling and upsampling components. Both visual results and quantitative evaluation metrics indicate that removing any of these network modules adversely affects the reconstruction performance of our network, demonstrating the effectiveness of the proposed model.

Table 5
Quantitative evaluation of ablation studies on three datasets, where the best-performing results are highlighted in bold black font (↓ denotes lower is better, ↑ denotes higher is better).

Dataset	Studied module	Angular Error ↓			Accuracy ↑		
		Mean	Median	RMSE	11.25°	22.5°	30.0°
DeepSfP	w/o I_{un}	15.71	12.06	20.13	46.95	77.40	88.06
	w/o ϕ_e	16.71	13.09	21.22	42.72	75.32	86.39
	w/o ρ	15.13	11.39	19.60	49.41	79.15	88.67
	w/o V	15.05	11.35	19.44	49.59	79.22	89.02
	w/o SC	15.14	11.43	19.52	49.27	78.89	88.56
	w/o CGA	15.55	11.79	20.01	47.92	77.65	87.94
	w/o Fusion	17.18	13.61	21.53	40.29	73.32	85.41
	w/o Transformer	15.42	11.69	19.82	48.27	78.33	88.57
	Ours-6.25% sampling	**14.95**	**11.13**	**19.40**	**50.46**	**79.32**	**88.70**
SPW	w/o I_{un}	25.66	19.08	34.64	40.12	66.25	74.45
	w/o ϕ_e	24.18	17.15	33.29	43.22	70.11	77.76
	w/o ρ	23.15	16.47	32.41	**45.77**	72.51	79.91
	w/o V	24.59	17.89	33.63	39.75	68.08	77.79
	w/o SC	25.41	18.12	34.62	39.93	68.54	76.17
	w/o CGA	25.82	19.73	34.32	35.91	64.87	74.65
	w/o Fusion	26.58	19.97	35.32	34.56	64.45	74.42
	w/o Transformer	23.26	16.68	32.37	45.29	71.88	79.71
	Ours-6.25% sampling	**19.66**	**14.62**	**26.43**	42.79	**73.03**	**81.74**
Real-world	w/o I_{un}	7.16	4.49	11.63	78.76	91.53	95.28
	w/o ϕ_e	7.50	5.26	11.64	78.92	91.73	95.38
	w/o ρ	7.52	3.50	12.58	74.30	89.95	94.57
	w/o V	8.19	4.02	13.51	71.24	88.36	93.77
	w/o SC	7.08	2.55	12.20	76.12	90.35	94.75
	w/o CGA	7.48	4.35	12.23	75.69	90.64	94.90
	w/o Fusion	7.57	**1.58**	13.05	73.63	89.01	94.03
	w/o Transformer	7.07	1.88	12.25	75.74	90.32	94.72
	Ours-6.25% sampling	**5.41**	1.67	**10.25**	**85.31**	**92.58**	**95.75**

5. Discussion

The above experimental results demonstrate that the four-quadrant parallel single-pixel ultra-low-sampling SfP imaging method based on decoupled spatial and polarimetric dimensions can maintain high surface normal reconstruction accuracy while substantially reducing the amount of acquired data. Experiments on the DeepSfP, SPW, and real-world datasets demonstrate the effectiveness of the proposed method under different sampling rates, in static scenes, and for targets with diverse material properties. In particular, at sampling rates of 1.56% and 6.25%, the system achieves reconstruction performance comparable to, and in some cases better than, several fully sampled deep-learning-based SfP methods with much lower data cost. These results indicate that incorporating the SPI undersampling mechanism into the SfP task can effectively reduce the storage, transmission, and backend processing burden of conventional SfP methods.

However, the proposed method still has several limitations. First, SPI requires multiple DMD coding patterns to be sequentially loaded while the target remains in the same state. Therefore, the end-to-end imaging time depends not only on the network inference speed, but also on the DMD pattern refresh rate, detector readout speed, number of measurements, and SPI physical decoding process. Although the proposed method significantly reduces the input data volume and transmission burden, the current prototype system cannot be regarded as a high-speed real-time SfP system. In particular, it remains difficult to meet the frame-rate requirements of high-speed industrial online inspection or dynamic augmented-reality scenarios. Second, when the target undergoes obvious motion or the ambient illumination changes rapidly, the consistency among different coded measurements can be disrupted, which further affects polarization-information estimation and surface normal recovery. Therefore, the current study mainly validates low-data 3D imaging under static-target conditions.

Future work will improve real-time performance and dynamic robustness from both hardware and algorithmic perspectives. On the hardware side, a higher-refresh-rate DMD or high-speed spatial light modulator could be used to shorten the sampling period for each frame, while synchronized exposure control and reference-channel normalization could reduce the influence of illumination fluctuations on the bucket signal. On the algorithmic side, adaptive sampling strategies

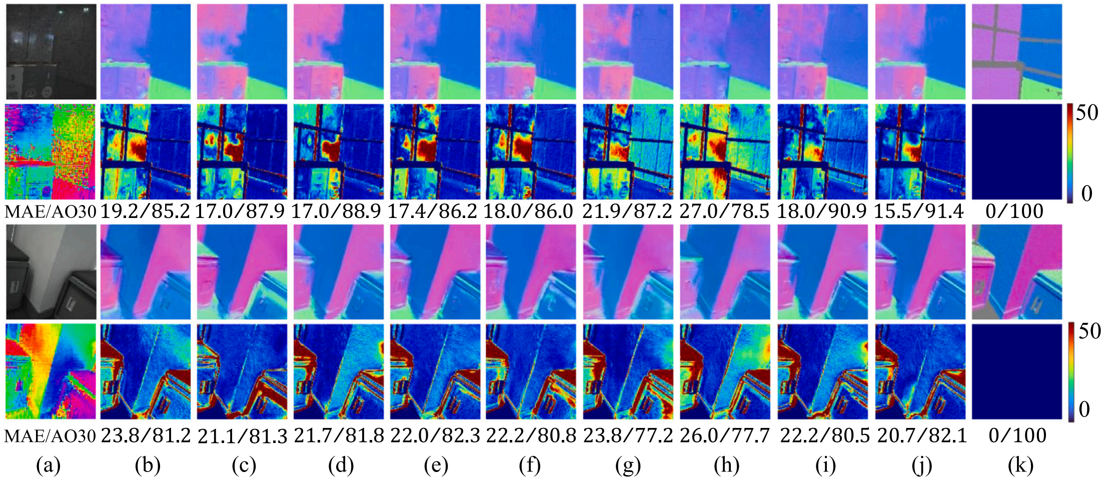


Fig. 11. Error analysis plot of the ablation experiment. MAE denotes the mean angular error, and AO30 represents the accuracy within 30°. (a) input and ϕ_e. (b) Without I_{un}. (c) Without ϕ_e. (d) Without ρ. (e) Without V. (f) Without SC. (g) Without CGA. (h) Without Fusion. (i) Without Transformer encoder. (j) Our-6.25% sampling rate. (k) the ground truth.

could be designed to dynamically select the sampling rate according to scene complexity. Meanwhile, lightweight end-to-end networks that directly recover surface normals from compressed measurements could be developed to reduce the computational cost introduced by intermediate SPI image reconstruction. In addition, deploying SPI decoding and deep-network inference on GPUs, FPGAs, or edge AI chips would further improve the system frame rate. With these improvements, the proposed method could be further extended to applications with higher real-time requirements, such as industrial online inspection, robotic navigation, and augmented reality.

6. Conclusion

This paper proposes an SPI ultra-low-sampling SfP system based on decoupled spatial and polarimetric dimensions, which synergistically integrates optical encoding and deep learning to achieve high-precision 3D imaging with significantly reduced data requirements. Experimental results demonstrate that the system can still preserve the main surface normal structure at the ultra-low sampling rate of 0.39%, while higher reconstruction accuracy is achieved at 1.56% and 6.25% sampling rates. The raw acquired data volume and equivalent transmission bandwidth are reduced by 256 times compared with the conventional full-sampling scheme of four polarization states. The proposed HyCGA-Transformer hybrid fusion framework effectively mitigates azimuthal π-ambiguity and zenith angle errors in polarization information through feature fusion and cross-scale semantic alignment, and demonstrates strong robustness in static scenes. Overall, this work overcomes key limitations of traditional SfP methods in data storage, transmission, and computational efficiency, while providing a lightweight and low-cost 3D sensing paradigm for static scenes under resource-constrained conditions, thereby promoting the development of SfP technology toward lightweight deployment.

CRediT authorship contribution statement

Baolin Wang: Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Data curation. **Yusong Liu:** Data curation. **Cheng Zhou:** Writing – review & editing, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Binyu Li:** Data curation. **Jipeng Huang:** Writing – review & editing, Methodology, Investigation, Funding acquisition, Formal analysis. **Dianlei Yao:** Data curation. **Qiyi Zhang:** Data curation. **Lijun Song:** Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data and code of this article have been disclosed on the project homepage.

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